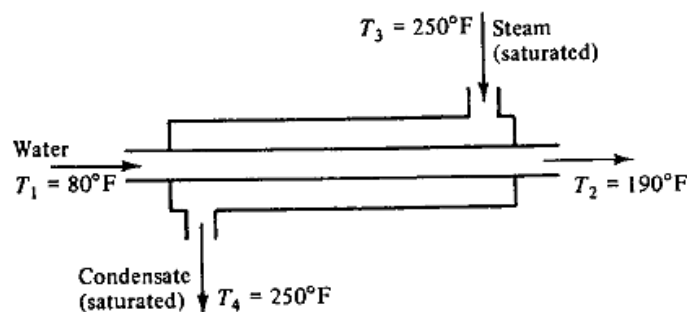


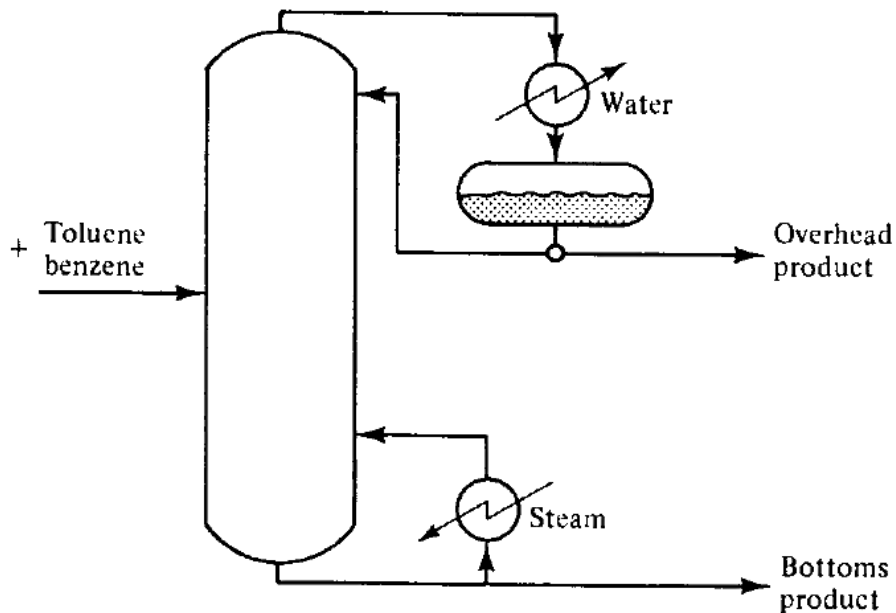
BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF CHEMICAL AND FOOD ENGINEERING
Chemical Engineering Program - BSc

Process Dynamics and Control - Worksheet 1

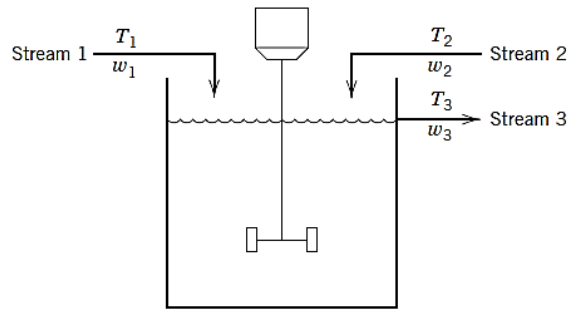
1. Consider the heat exchanger shown below and you are appointed as a control engineer to design the control system for it. Identify:
 - a. The control objectives for the system
 - b. All the external disturbances that will affect the operation of the system
 - c. All the available manipulated variables for the control of the system in the presence of disturbances.
 - d. Construct all the possible feedback and feedforward control configurations that will satisfy the control objectives in the presence of disturbances.



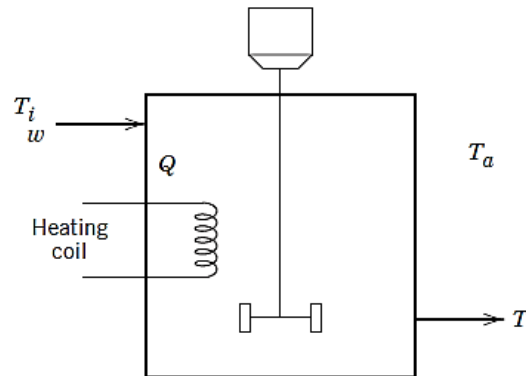
2. The distillation column shown below is designed to separate toluene from benzene. Identify:
 - a. All the control objectives, all external disturbances, and all the available measurements and manipulated variables.
 - b. Construct feedback and feedforward controller configurations to account for changes in the feed flow rate.
 - c. Develop an inferential control system to keep the purity of benzene.



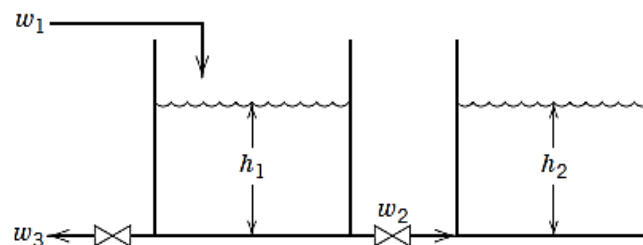
3. Consider a system of two continuous stirred tank reactors in series where the reactant (A) and catalytically converted to the product (B) in an endothermic reaction.
 - a. Identify the control objectives for the operation of the system.
 - b. Classify the variables of the system into inputs, outputs (measured and unmeasured), disturbances, manipulated variables.
 - c. Develop feedback and feedforward control configurations that satisfy your control objectives using a composition analyzer when needed.
 - d. Develop an inferential control configuration assuming that a composition analyzer is not available.
 - e. Compare the control configuration you developed in c and d and select the best one to achieve the same conversion.
4. A perfectly stirred, constant-volume tank has two input streams, both consisting of the same liquid (figure below). The temperature and flow rate of each of the streams can vary with time. w_i denotes mass flow rate for stream i . Develop a dynamic model that will describe the transient operation of the system and make a degrees of freedom analysis assuming that both Streams 1 and 2 come from upstream units (i.e., their flow rates and temperatures are known functions of time). How do you make the problem a completely described one?



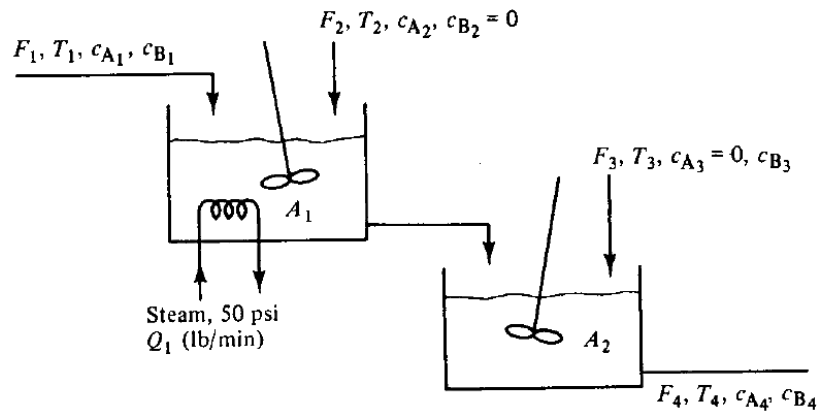
5. A completely enclosed stirred-tank heating process shown below is used to heat an incoming stream whose flow rate varies. The heating rate from the coil and the volume of the tank are both constant. Develop a mathematical model that describes the exit temperature if heat losses to the ambient occur and if the ambient temperature (T_a) and the incoming stream's temperature (T_i) both can vary. Discuss qualitatively what happens to the exit temperature as T_i and w increase (or decrease) and justify it by reference to your model.



6. Two tanks are connected together as shown in figure below. Making all the necessary assumptions, develop mathematical model to find h_1 , h_2 , w_2 , and w_3 as functions of time for any given variations in inputs. Perform a degrees of freedom analysis. Identify all input and output variables.



7. Consider the mixing process taking place in a two-tank system as shown below. Identify the state variables and develop the mathematical model that describes the dynamic behaviour of the process assuming that the heats of solution are strong functions of the composition.



8. A closed stirred-tank reactor with two compartments is shown in figure below. The basic idea is to feed the reactants continuously into the first compartment, where they will be preheated by energy liberated in the exothermic reaction, which is anticipated to occur primarily in the second compartment. The wall separating the two compartments is quite thin, thus allowing heat transfer; the outside of the reactor is well insulated; and a cooling coil is built into the second compartment to remove excess energy liberated in the reaction. Both tanks are completely full at all times. Identify the control objectives and develop a dynamic model for the process assuming that reactant A is converted to product B. Make a degrees of freedom analysis for your model, identifying all parameters, outputs, and inputs that must be known functions of time in order to obtain a solution.